Journal of the Optical Society of America B
OPTICAL PHYSICS

Compact magneto-optical trap with a quartz vacuum chamber for miniature gravimeters

KANXING WENG,[1,†] BIN WU,[1,†] JIAHONG LIN,[1] YIN ZHOU,[1] BING CHENG,[1,2] AND QIANG LIN[1,3]

[1] Institute of Optics, The Zhejiang Provincial Key Laboratory of Quantum Precision Measurement, College of Science, Zhejiang University of Technology, Hangzhou 310023, China
[2] e-mail: bingcheng@zjut.edu.cn
[3] e-mail: qlin@zjut.edu.cn

Received 13 February 2020; revised 22 March 2020; accepted 7 April 2020; posted 8 April 2020 (Doc. ID 390512); published 0 MONTH 0000

The realization of a compact magneto-optical trap (MOT) with a quartz vacuum chamber via ultra-low power of a cooling laser is reported. The dependence of the properties of the MOT on key experimental parameters is investigated. The efficiency of capturing atoms of the novel MOT is compared with that of the metal vacuum chamber. The compact quartz vacuum chamber can achieve a maximum rubidium (^{87}Rb) atomic loading rate of about 8×10^9/s with less than 60 mW of the total power of cooling lasers, while the metal chamber obtains the maximum atomic loading rate of about 3×10^9/s with more than 240 mW of the total power of cooling lasers. The compact apparatus avoids undesired effects found in a metal chamber, such as high power consumption and the magnetic-field eddy current effect. This compact MOT is helpful in the design of miniature atomic gravimeters.

https://doi.org/10.1364/JOSAB.390512

1. INTRODUCTION

The cold atom gravimeter developed in recent decades with ultra-high sensitivity has shown its importance for precision measurement [1–6]. The preparation of a sufficient amount of cold atoms is the basic process of the atom gravimeter [7–9], and it is the same for quantum information [10] and quantum simulation [11,12]. Meanwhile, the invention and utilization of magneto-optical trap (MOT) technology have greatly simplified the technology of capturing cold atoms. So far, there have been many excellent reviews on the basic theories of MOTs [13–16]. With the progress of MOT technology, the structure of MOTs has become diversified, from simple three-dimensional MOTs (3D-MOTs) to two-dimensional MOTs (2D-MOT) combined with 3D-MOTs [17], as well as the latest grating MOT (G-MOT) [18], pyramid MOT (P-MOT) [19], and mirror MOT (M-MOT) [20]. Among all these methods, the 2D-MOTs are used to produce a continuous slow atom flux [21–23], which can effectively increase the loading rate and atomic number in 3D-MOTs. It has become standard MOT technology for trapping atoms and molecules. However, most of the vacuum chambers of MOTs are made of metal, which is bulky and unsuitable for miniature atomic sensors. In addition, they often need the strong power of a cooling laser, and it is hard to avoid the magnetic-field eddy current effect [24–27].

Therefore, some research groups have shown interest in investigating quartz vacuum chambers. Altin *et al.* [28] trapped approximately 10^8 atoms (loading cycle of 500 ms) via a custom-built glass octagon. Butts *et al.* [29] trapped cesium from a background vapor with an octagonal glass cell.

In this paper, we have realized the maximal ^{87}Rb atomic loading rate of about 8×10^9/s in the compact quartz vacuum chamber, and only less than 60 mW of the total power of cooling lasers is needed. In addition, we compare it with the metal chamber, and it obtains the maximum atomic loading rate of about 3×10^9/s with more than 240 mW of the total power of cooling lasers. Due to the compact, elegant design of the vacuum chamber, it simplifies obtaining a sufficient magnetic-field gradient and laser power to run the MOT. It is demonstrated in this paper that the compact quartz vacuum chamber has the ability of ultra-low power consumption compared with the metal chamber.

2. APPARATUS

A. Vacuum Chamber

The schematic of the compact quartz vacuum chamber is sketched in Fig. 1(a). The material of the vacuum chamber system is quartz, and the mass of the chamber is less than 620 g. It is constructed through the technology of hydroxide catalyzed bonding with many advantages, such as high transmittance (both sides of the quartz are AR coated), low coefficient of thermal expansion, lack of magnetic field, and low helium permeability. Therefore, it can effectively avoid the system error caused by the magnetic-field eddy current and other effects

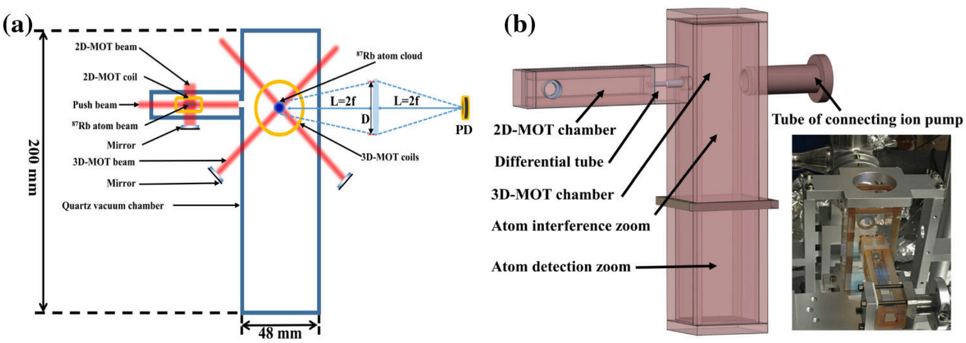

Fig. 1. Schematic of the experimental setup of our chamber. (a) Schematic diagram of the compact quartz vacuum chamber: atomic loading rate is detected by PD; detected fluorescence light from the trapped atoms; $f = 50$ mm is the focal length of the lens, $D = 12.5$ mm is the diameter of the lens, $L = 10$ cm is the distance between lens and atoms, and PD is the silicon photodetector. (b) Technical drawing of the compact quartz vacuum chamber.

compared with metal [27]. To maintain low pressure in the 3D-MOT chamber, the rubidium source is placed in the 2D-MOT and transfers the atoms via the differential tube. The 3D-MOT chamber maintains the background pressure of about 1.35×10^{-9} Torr with a 5 L/s ion pump, which connects with the chamber by crimping the indium wire (not shown in Fig. 1).

To increase the atomic loading rate and the number of atoms, the 2D-MOT is used for pre-trapping atoms. The optical setup of the 2D-MOT is composed of two pairs of counter-propagating, circularly polarized cooling lasers, and the lasers orthogonally cross at the zero of a magnetic quadrupole field. The 3D-MOT system is equipped with three pairs of counter-propagating, circularly polarized cooling lasers, where the lasers cross at the zero of a magnetic quadrupole field and are used for trapping and cooling the atoms from the 2D-MOT.

The axial magnetic-field gradient of the 2D-MOT is provided by two pairs of orthogonal ($27\,\text{mm} \times 13\,\text{mm}$) anti-Helmholtz coils, where the shape of the 2D-MOT coils is rectangle with 88 turns. Moreover, the axial magnetic-field gradient of the 3D-MOT is provided by a pair anti-Helmholtz coils, where the 3D-MOT coils have 138 turns, and the radius of the coils is 20 mm. In addition, all the MOT coils are placed outside and close to the chamber, which can avoid contamination of the interior of the chamber and is easy to replacement.

B. Laser System

The laser system has been described elsewhere [30]. As shown in Fig. 2, the laser system is based on two 1560 nm distributed feedback (DFB) laser diodes. We utilized one as the master laser and the other one as the reference laser. The frequency of the reference laser is multiplied by the periodically poled lithium niobate waveguide (PPLN-WG) and locked to the ^{85}Rb transition of the D2 line via the method of modulation transfer spectroscopy (MTS). The master laser is locked via the beat note with the frequency-locked laser, the power is amplified by an erbium-doped fiber amplifier (EDFA), and the frequency is doubled by the PPLN crystal. We used digital frequency-hopping locking (DFHL) technology to reinforce the function of the phase-locked loop based on traditional optical phase locking.

The cooling laser is transmitted to the MOT via polarization-maintaining (PM) fibers and expanded by the air-spaced

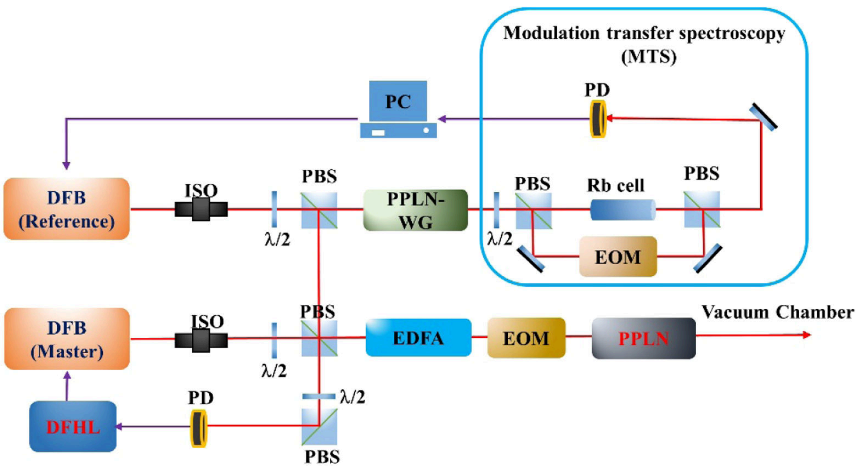

Fig. 2. Schematic diagram of the laser system for ^{87}Rb MOT. DFB, distributed feedback laser; ISO, isolator; $\lambda/2$, half-wave plate; PBS, polarizing laser splitter; PPLN-WG, periodically poled lithium niobate waveguide; EOM, electro-optic modulator; EDFA, erbium-doped fiber amplifier; Rb cell, rubidium vapor cell; PD, photodiode; DFHL, digital frequency-hopping locking.

doublet collimator (Thorlabs F810APC-780) with a $1/e^2$ waist of 7.5 mm. The 3D-MOT lasers will maintain the laser spot, and the 2D-MOT lasers will be further expanded to the size of 5 mm × 25 mm.

3. EXPERIMENTAL RESULTS

A. Properties of the 3D-MOT

In order to investigate the characteristics of the MOT with a quartz chamber, it is necessary to study how to maximize the loading rate of the atoms—first, by adjusting the 2D-MOT's cooling lasers to ensure that the horizontal and vertical lasers are orthogonal at the center of the magnetic field of the 2D-MOT. In fact, only with good alignment and parameters (power, polarization, detuning, and magnetic-field gradient) can we visually see the cold atom in the 2D-MOT via a CCD camera, where the thin bright line in Fig. 3(a) shows the photo of the atomic beam in the 2D-MOT. Second, the atomic beam will be transferred to the 3D-MOT via the push laser. In order to load the atoms efficiently, it is important to optimize the properties of the 3D-MOT cooling lasers to ensure that as many atoms as possible are trapped, and we can observe a bright spot in the center of the 3D-MOT via the CCD, as shown in Fig. 3(b). Finally, we measure the fluorescence signal of the trapped atoms in the 3D-MOT via the silicon photodetector (PD), which has been placed against the center of the 3D-MOT (see Fig. 1).

The fluorescence signal makes a good measurement of atomic loading rate, and it can be utilized to characterize the properties of the 3D-MOT. By optimizing the key parameters of the MOT with a quartz (metal) chamber, we can obtain the maximum fluorescence signal of each chamber. The red (black) curve in Fig. 4 shows the atomic loading process with the quartz (metal) chamber, and the loading process lasts about 200 ms (300 ms) until it reaches saturation. Thus, the difference in performance between the two chambers can be clearly observed, where the fluorescence intensity of the quartz chamber is significantly better than that of the metal chamber, and the loading time of the quartz chamber is only 200 ms, which is shorter than the 300 ms of the metal chamber.

Then, the atomic loading rate can be calculated by the following equations [31]. The result shows that the maximum loading rate of the compact quartz chamber is 8×10^9/s, and the maximum loading rate of the metal chamber is 3×10^9/s:

$$P_d = \frac{U_{\mathrm{Flu}}}{R_{es}\eta_0}, \tag{1}$$

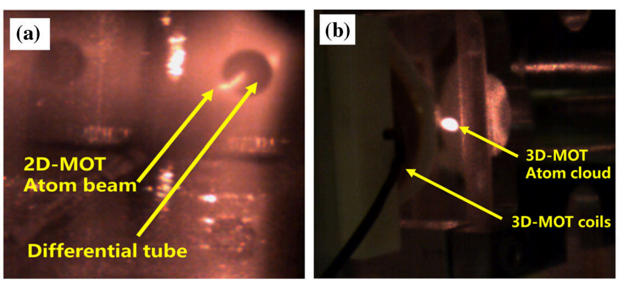

Fig. 3. Photographs of cold atom in 2D-MOT and 3D-MOT via CCD camera. (a) 2D-MOT atom beam and (b) 3D-MOT atom cloud.

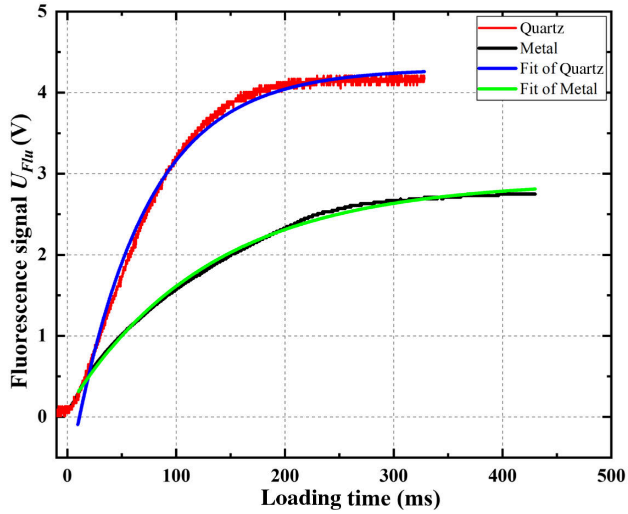

Fig. 4. Curves of atomic loading. The red and black curves are the fluorescence signal of the atomic loading with quartz chamber and metal chamber, respectively. The blue and green curves are exponential fits to Eq. (3).

where P_d is the fluorescence intensity received by the PD (shown in Fig. 2), $R_{es} = 10\ \mathrm{M\Omega}$ is the sampling resistor, $\eta_0 = 0.60$ A/W is the photo-sensitivity of the PD, and U_{Flu} is the voltage signal measured from the fluorescence signal:

$$N_{\mathrm{atoms}} = \frac{64 P_d L^2(\delta^2 + 0.25\Gamma^2 + 0.12\Gamma^2 I_0/I_{\mathrm{sat}})}{0.23\eta_1\eta_2 D^2\hbar\omega\Gamma^3 I_0/I_{\mathrm{sat}}}, \tag{2}$$

where N_{atoms} is the steady state number of the atoms, I_0 is the intensity of the 3D-MOT per cooling laser, I_{sat} is the saturation intensity of the cooling transition, η_1 is the vacuum chamber transmittance, η_2 is the lens transmittance, $D = 12$ mm is the diameter of the lens, $\hbar\omega$ is the transition energy; $L = 10$ cm is the distance between the lens and atoms, δ is the detuning between the $5^2S_{1/2}$, $F = 2 \to 5^2P_{1/2}$, $F' = 3$ resonance frequencies, and $\Gamma \approx 2\pi * 6$ MHz is the natural line width of the ^{87}Rb D2 line:

$$N(t) = N_{\mathrm{atoms}}[1 - \exp(-t/\tau)], \tag{3}$$

$$R = \frac{N_{\mathrm{atoms}}}{\tau}, \tag{4}$$

where R is the atomic loading rate and τ is the atomic lifetime.

We first investigated the atomic loading rate versus the power of the cooling laser in the 3D-MOT with a quartz chamber. The filled points in Fig. 5(a) show how the loading rate depends on the power of I_{3D}. The loading rate increases roughly linearly with I_{3D} until I_{3D} reaches maximum 7 mW. The filled points in Fig. 5(b) are the results of the loading rate depending on the detuning δ, and the δ controlled by the DFHL (shown in Fig. 2). For the detuning δ below −32 MHz, the 3D-MOT cannot trap atoms, and it is the same for the δ above −5 MHz. Beyond this range, the loading rate increases with δ until it reaches the peak at a detuning near −15 MHz; then it will decrease with δ. Thus, we can deduce the optimal experimental parameters of $I_{3D} = 7$ mW and $\delta = -15$ MHz.

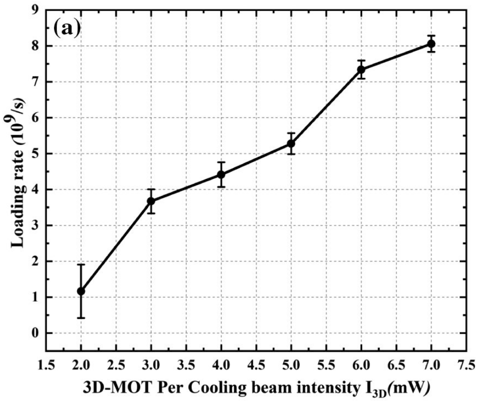

Fig. 5. Characteristic of the atomic loading rate versus the I_{3D} and δ. (a) Atomic loading rate versus the I_{3D}, and (b) atomic loading rate versus the δ.

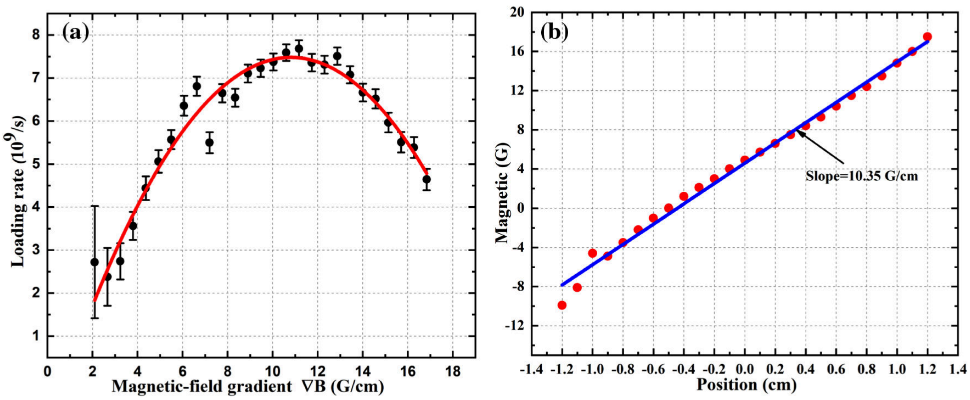

Fig. 6. Characteristic of the atomic loading rate versus the magnetic-field gradient and the measurement of magnetic field in 3D-MOT. (a) The black points represent the atomic loading rate depending on the ∇B in the MOT, and the red curve is fits of parabolic function; (b) the red circles are the magnetic field versus the positions of the 3D-MOT via the magnetometer. The blue line with a slope of 10.35 Gauss/cm is the fit of the axial magnetic-field gradient.

Next we investigated the atomic loading rate versus the axial magnetic-field gradient ∇B of the 3D-MOT in a quartz chamber. The axial magnetic-field gradient is produced by a pair of anti-Helmholtz coils placed outside and near the center of the MOT chamber, as illustrated in Fig. 1, and the axial magnetic-field gradient can be controlled by changing the current of the coils. The filled points in Fig. 6(a) show how the loading rate depends on the axial magnetic-field gradient, and the trend is similar to a parabolic function with a maximum at $\nabla B = 10$ Gauss/cm. For the axial magnetic-field gradient below 10 Gauss/cm, the loading rate increases roughly linearly, and the loading rate decreases linearly with the magnetic-field gradient above 10 Gauss/cm, so we deduce the optimal magnetic-field gradient of about 10 Gauss/cm. Next, setting the current of 3D-MOT coils at 2 A produces the axial magnetic-field gradient ∇B near 10 Gauss/cm, as shown in Fig. 6(b). The measurement of the axial magnetic-field gradient of the MOTs via the magnetometer and the position of 0 cm represent the center of the Helmholtz coils. Thus, using a suitable magnetic-field gradient can not only increase the atomic loading rate, but also effectively reduce the uncertain effects of thermal radiation from the coils.

B. Comparison of Metal Chamber and Quartz Chamber

To compare the difference in atomic loading efficiency between quartz and metal chambers, we configured the optimal parameters of the 3D-MOT with a quartz (metal) chamber, where $I_{3D} = 7(7)$ mW, $\delta = -15(-12)$ MHz, and $\nabla B = 10(10)$ Gauss/cm to complete the contrast experiment of a metal chamber and quartz chamber. Since the same laser system is used, the optimal parameters of each chamber are relatively close. In addition, the metal chamber is made of titanium, and it maintains background pressure of about 1.12×10^{-9} Torr with a 40 L/s ion pump.

Figure 7 shows how the loading rate depends on the power of I_{2D}. The red points in Fig. 7(a) are the result of the loading rate versus the I_{2D} in the MOT with the metal chamber. For the I_{2D} below 25 mW, the fluorescence signal of the atom is too low to observe, while above this threshold, the loading rate increases roughly linearly with I_{2D} until it reaches a peak of about 3×10^9/s near $I_{2D} = 100$ mW. However, the loading rate will fall above $I_{2D} = 100$ mW, because at high power, the atoms decelerate too fast. The blue points in Fig. 7(b) are the result of the loading rate depending on the I_{2D} with a compact

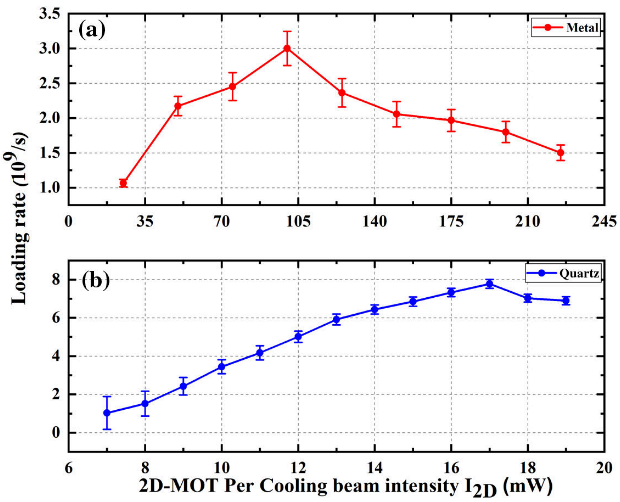

Fig. 7. Characteristic of the atomic loading rate versus the I_{2D}, which holds the other optimal parameters. (a) The red points represent the atomic loading rate depending on the I_{2D} in the MOT with metal chamber, and (b) the blue points represent the atomic loading rate depending on the I_{2D} in the MOT with quartz chamber.

quartz chamber MOT. For the I_{2D} below 7 mW, the loading rate of atoms is too low to measure, and above this threshold, the loading rate increases as a linear function until it reaches a peak of about 8×10^9/s near $I_{2D} = 17$ mW, and the loading rate will fall above $I_{2D} = 17$ mW.

Therefore, the optimal parameter is $I_{2D} = 100$ mW for the metal chamber, and it can obtain a maximal loading rate of about 3×10^9/s. Compared to the metal chamber with the optimal parameter of I_{2D}, the quartz chamber needs only 17 mW. However, the maximal loading rate of the quartz chamber can obtain about 8×10^9/s. Thus, we can conclude that with the optimal parameters of the 3D-MOT, the compact quartz chamber has better capture efficiency. Due to the compact structure, the distance between our 2D-MOT and 3D-MOT is short enough to reduce the atomic loss in the process of atoms transferred from 2D-MOT to 3D-MOT.

In summary, we have investigated how the properties of the 3D-MOT depend on the key parameters of the compact quartz chamber by implementing fluorescence signal measurement, including the power of a 3D-MOT cooling laser I_{3D}, the detuning δ, and the axial magnetic-field gradient ∇B. Through a series of experiments, we found a set of optimal parameters $I_{3D} = 7$ mW, $\delta = -15$ MHz, and $\nabla B = 10$ Gauss/cm for our apparatus. Then, we used these optimal parameters to compare with a metal chamber and found the optimal parameter of the power of a 2D-MOT cooling laser $I_{2D} = 17$ mW (100 mW) of the quartz chamber (metal chamber). The results show that the quartz chamber has better atomic loading efficiency than the metal chamber, and is more suitable for miniature atomic gravimeters.

4. CONCLUSION

We have realized a MOT with a quartz chamber, which is compact, light, and easy to use for miniature atomic gravimeters. The properties of the 3D-MOT versus the key parameters (I_{3D}, δ, ∇B, I_{2D}) on the atomic loading rate have been investigated by implementing fluorescence signal measurement, and we have derived a set of optimal parameters for our compact quartz chamber ($I_{3D} = 7$ mW, $I_{2D} = 17$ mW, $\delta = -15$ MHz and $\nabla B = 10$ Gauss/cm). By setting these parameters of the MOT, it can load atoms with a maximum loading rate of about 8×10^9/s, and the total power of the cooling laser is less than 60 mW.

By gradually increasing the power of 3D-MOT per cooling laser I_{3D}, the loading rate of the atoms increases linearly. Thus, one possible way to increase the maximum loading rate of the MOT is by appropriately increasing the I_{3D}. The detuning δ versus the atomic loading rate is similar to a parabolic function with a maximum at $\delta = -15$ MHz, and this value may fluctuate depending on the line width of the laser. The loading rate depends on the axial magnetic-field gradient ∇B increasing as a parabolic function with a maximum at $\nabla B = 10$ Gauss/cm. Benefiting from the compact design, the MOT generates a sufficient magnetic-field gradient with only 10 W of power (the current of MOT coils about 2 A). As the gradient is increased beyond this, there is a slow decline in the number of trapped atoms, suggesting that the trap capture volume starts to decrease at these higher gradients [32]. We configured the optimal parameters of each chamber to compare the loading rate and the total power of the cooling laser. The results show that the compact quartz chamber has better capture efficiency, due to the quartz chamber being compact and the length of the differential tube between 2D-MOT and 3D-MOT being short enough to reduce the atomic loss in the process of atoms from 2D-MOT to 3D-MOT. Actually, the push laser is the source of heating the cold atoms. The longer the transport distance, the higher the atomic velocity, and it is not easy to be captured via the 3D-MOT. In addition, the quartz vacuum chamber can avoid the effects caused by the magnetic-field eddy current effect. Although we cannot accurately evaluate the impact for the time being, the residual magnetic field will certainly destroy the background magnetic field of the MOT and bring uncertain system errors.

Moreover, the difficulty of obtaining and maintaining the vacuum of the quartz chamber is the preparation and installation of the chamber. The quartz chamber is prepared by the technology of hydroxide-catalyzed bonding, and the chamber is fragile. Therefore, sufficient experience and means are required to ensure preparation and installation success. In contrast, the difficulty of the metal chamber is the crimping of the metal and the light-transmitting windows. The uneven crimping can cause the windows to deform or even crack.

Finally, the atomic loading rate could be further enhanced by increasing the temperature of the atom source. For example, the melting point of rubidium is 39°C, and heating the rubidium source to 40°C can effectively enhance the loading rate.

Funding. National Key Research and Development Program of China (2016YFF0200206, 2017YFC0601602); National Natural Science Foundation of China (11404286, 51905482, 61727821, 61875175); Research funds from the Ministry of Natural Resources of China (DD20189831).

Disclosures. The authors declare no conflicts of interest.

†These authors contributed equally to this work.

REFERENCES

1. S. Chu, J. E. Bjorkholm, A. Ashkin, and A. Cable, "Experimental observation of optically trapped atoms," Phys. Rev. Lett. **57**, 314–317 (1986).
2. M. Kasevich and S. Chu, "Atomic interferometry using stimulated Raman transitions," Phys. Rev. Lett. **67**, 181–184 (1991).
3. M. Kasevich, D. S. Weiss, E. Riis, K. Moler, S. Kasapi, and S. Chu, "Atomic velocity selection using stimulated raman transitions," Phys. Rev. Lett. **66**, 2297–2300 (1991).
4. K. Moler, D. S. Weiss, M. Kasevich, and S. Chu, "Theoretical analysis of velocity-selective Raman transitions," Phys. Rev. A **45**, 342–348 (1992).
5. M. Kasevich and S. Chu, "Measurement of the gravitational acceleration of an atom with a light-pulse atom interferometer," Appl. Phys. B **54**, 321–332 (1992).
6. R. Caldani, K. X. Weng, S. Merlet, and F. Pereira Dos Santos, "Simultaneous accurate determination of both gravity and its vertical gradient," Phys. Rev. A **99**, 033601 (2019).
7. K. K. Ni, S. Ospelkaus, M. H. G. de Miranda, A. Pe'er, B. Neyenhuis, J. J. Zirbel, S. Kotochigova, P. S. Julienne, D. S. Jin, and J. Ye, "A high phase-space-density gas of polar molecules," Science **322**, 231–235 (2008).
8. B. H. McGuyer, M. McDonald, G. Z. Iwata, M. G. Tarallo, W. Skomorowski, R. Moszynski, and T. Zelevinsky, "Precise study of asymptotic physics with subradiant ultracold molecules," Nat. Phys. **11**, 32–36 (2015).
9. M. Xiaobin, Y. Zhuxiong, X. Liyang, G. Zhen, Y. Li, and T. Meng Khoon, "A dual-species magneto-optical trap for lithium and strontium atoms," arXiv:1904.05230 (2019).
10. A. V. Gorshkov, A. M. Rey, A. J. Daley, M. M. Boyd, J. Ye, P. Zoller, and M. D. Lukin, "Alkaline-earth-metal atoms as few-qubit quantum registers," Phys. Rev. Lett. **102**, 110503 (2009).
11. M. Hofer, L. Riegger, F. Scazza, C. Hofrichter, D. R. Fernandes, M. M. Parish, J. Levinsen, I. Bloch, and S. Folling, "Observation of an orbital interaction-induced Feshbach resonance in ^{173}Yb," Phys. Rev. Lett. **115**, 265302 (2015).
12. G. Pagano, M. Mancini, G. Cappellini, L. Livi, C. Sias, J. Catani, M. Inguscio, and L. Fallani, "Strongly interacting gas of two-electron fermions at an orbital Feshbach resonance," Phys. Rev. Lett. **115**, 265301 (2015).
13. S. P. Xu, M. Xia, R. X. Gu, Y. N. Yin, L. Xu, Y. Xia, and J. P. Yin, "Three-dimensional modeling of magneto-optical trapping of MgF molecules with multilevel rate equations," Phys. Rev. A **99**, 033408 (2019).
14. W. Suptitz, G. Wokurka, F. Strauch, P. Kohns, and W. Ertmer, "Simultaneous cooling and trapping of ^{85}Rb and ^{87}Rb in a magneto-optical trap," Opt. Lett. **19**, 1571–1573 (1994).
15. F. M. Peixoto, "Enhanced loading of a lithium-7 magneto optical trap using transverse cooling and frequency spread light," Ph.D. thesis (Yale University, 2002).
16. T. G. Tiecke, S. D. Gensemer, A. Ludewig, and J. T. M. Walraven, "High-flux two-dimensional magneto-optical-trap source for cold lithium atoms," Phys. Rev. Lett. **80**, 013409 (2009).
17. E. L. Raab, M. Prentiss, A. Cable, S. Chu, and D. E. Pritchard, "Trapping of neutral sodium atoms with radiation pressure," Phys. Rev. Lett. **59**, 2631–2634 (1987).
18. J. Lee, J. A. Grover, L. A. Orozco, and S. L. Rolston, "Sub-Doppler cooling of neutral atoms in a grating magneto-optical trap," J. Opt. Soc. Am. B **30**, 2869–2874 (2013).
19. W. Bowden, R. Hobson, I. R. Hill, A. Vianello, M. Schioppo, A. Silva, H. S. Margolis, P. E. G. Baird, and P. Gill, "A pyramid MOT with integrated optical cavities as a cold atom platform for an optical lattice clock," Sci. Rep. **9**, 9 (2019).
20. M. Hyodo, K. Nakayama, M. Watanabe, and R. Ohmukai, "Mirror magneto-optical trap exploiting hexapole-compensated magnetic field," Phys. Rev. A **76**, 013419 (2007).
21. I. Nosske, L. Couturier, F. C. Hu, C. Z. Tan, C. Qiao, J. Blume, Y. H. Jiang, P. Chen, and M. Weidemuller, "Two-dimensional magneto-optical trap as a source for cold strontium atoms," Phys. Rev. A **96**, 053415 (2017).
22. Z. T. Lu, K. L. Corwin, M. J. Renn, M. H. Anderson, E. A. Cornell, and C. E. Wieman, "Low-velocity intense source of atoms from a magneto-optical trap," Phys. Rev. Lett. **77**, 3331–3334 (1996).
23. K. Dieckmann, R. J. C. Spreeuw, M. Weidemuller, and J. T. M. Walraven, "Two-dimensional magneto-optical trap as a source of slow atoms," Phys. Rev. Lett. **58**, 3891–3895 (1998).
24. X. W. Zhang, J. Q. Zhong, B. A. Tang, X. Chen, L. Zhu, P. W. Huang, J. Wang, and M. S. Zhan, "Compact portable laser system for mobile cold atom gravimeters," Appl. Opt. **57**, 6545–6551 (2018).
25. X.-L. Wang, B. Cheng, B. Wu, Z.-Y. Wang, and Q. Lin, "A simplified cold atom source for 3-D MOT loading," Chin. Phys. Lett. **28**, 053701 (2011).
26. Q. Y. Wang, Z. Y. Wang, Z. J. Fu, and Q. Lin, "Investigation of the thermal adaptability for a mobile cold atom gravimeter," Chin. Phys. B **25**, 123701 (2016).
27. B. Wu, Z. Y. Wang, B. Cheng, Q. Y. Wang, A. P. Xu, and Q. Lin, "The investigation of a mu Gal-level cold atom gravimeter for field applications," Metrologia **51**, 452–458 (2014).
28. P. A. Altin, M. T. Johnsson, V. Negnevitsky, G. R. Dennis, R. P. Anderson, J. E. Debs, S. S. Szigeti, K. S. Hardman, S. Bennetts, G. D. McDonald, L. D. Turner, J. D. Close, and N. P. Robins, "Precision atomic gravimeter based on Bragg diffraction," New J. Phys. **15**, 023009 (2013).
29. D. L. Butts, J. M. Kinast, B. P. Timmons, and R. E. Stoner, "Light pulse atom interferometry at short interrogation times," J. Opt. Soc. Am. B **28**, 416–421 (2011).
30. Q. Y. Wang, Z. Y. Wang, Z. J. Fu, W. Y. Liu, and Q. Lin, "A compact laser system for the cold atom gravimeter," Opt. Commun. **358**, 82–87 (2016).
31. C. Monroe, H. Robinson, and C. Wieman, "Observation of the cesium clock transition using laser-cooled atoms in a vapor cell," Opt. Lett. **16**, 50–52 (1991).
32. H. J. Williams, S. Truppe, M. Hambach, L. Caldwell, N. J. Fitch, E. A. Hinds, B. E. Sauer, and M. R. Tarbutt, "Characteristics of a magneto-optical trap of molecules," New J. Phys. **19**, 113035 (2017).